

High Level Design (HLD) - RAG Customer Support System

1. System Overview

Problem: Traditional chatbots fail in domain-specific accuracy and hallucinate answers.

Scope: Build a RAG-based customer support system with HITL and memory using PDF knowledge base.

2. Architecture Overview

User Interface (CLI/Web) → Document Ingestion → Embedding Layer → Vector DB (ChromaDB) → Retriever → LLM (Groq) → LangGraph Workflow → HITL System.

3. Components

Document Loader: Loads PDF data.

Chunking Strategy: Splits text into 800-token chunks with overlap.

Embedding Model: HuggingFace embeddings.

Vector Store: ChromaDB.

Retriever: Semantic similarity search.

LLM: Groq LLaMA 3.

Graph Engine: LangGraph.

HITL Module: Escalates uncertain queries.

4. Data Flow

PDF → Chunking → Embeddings → Vector DB → Query → Retrieval → Context → LLM → Answer / Escalation.

5. Technology Choices

ChromaDB: Lightweight local vector store.

LangGraph: Stateful workflow orchestration.

Groq LLM: Fast inference.

HuggingFace embeddings: Efficient semantic representation.

6. Scalability

Supports large PDFs via chunking.

Optimized retrieval reduces latency.

Can scale via distributed vector DB and API deployment.

